

# KUMAR SAMBHAV

Software Engineer — Backend & AI/ML

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— [LinkedIn](#) — [Portfolio](#)

## Education

**SRM Institute of Science and Technology, Chennai** 2023 – 2027  
*B.Tech, Computer Science Engineering — Minor: Financial Technology (FinTech) — CGPA: 9.01/10.00*  
DAV Public School, Jamshedpur — Class XII (CBSE) 74% — 2023  
Sheyn International School, Jamshedpur — Class X (CBSE) 79% — 2021

## Technical Skills

**Languages:** Python, Java, C, C++, JavaScript, TypeScript  
**Frontend:** React.js, Next.js, Tailwind CSS, HTML5, CSS3  
**Backend:** FastAPI, Flask, Django, Node.js, Express.js  
**Databases:** PostgreSQL, MySQL, MongoDB  
**Machine Learning:** TensorFlow, Scikit-Learn, Pandas, NumPy, OpenCV, Matplotlib  
**DevOps & Tools:** Docker, Kubernetes, Prometheus, Grafana, Git, GitHub Actions, Linux, VS Code  
**Languages Spoken:** English, Hindi, Basic German

## Projects

**HealStack AI — Self-Healing Infrastructure Platform** *FastAPI, PostgreSQL, Docker, Prometheus, Grafana, Python*

- Architected an autonomous infrastructure monitoring system that ingests real-time metrics and logs, applying anomaly detection algorithms to flag incidents before they escalate into outages.
- Designed automated root-cause analysis and remediation workflows that diagnose failures and trigger self-healing actions, reducing manual on-call intervention and mean time to recovery (MTTR).

**DeepTrace AI — Deepfake Detection & Media Forensics Platform** *TensorFlow, OpenCV, FastAPI, Python*

- Built a multi-modal deepfake detection pipeline combining CNN-based image classifiers and frame-level video analysis to identify manipulated media with high accuracy.
- Implemented an explainable AI reporting layer that surfaces forensic evidence (artifact heatmaps, confidence scores) via a FastAPI service for transparent, auditable predictions.

**AstraQuant AI — Financial Intelligence Platform** *Python, FastAPI, PostgreSQL*

- Developed a financial analytics engine that aggregates market data and computes risk metrics and portfolio insights through a RESTful FastAPI backend.
- Designed a PostgreSQL schema and AI-driven prediction module to generate data-backed investment signals, enabling scalable portfolio intelligence workflows.

**Cotton Plant Disease Detection — Deep Learning Classifier** *TensorFlow, OpenCV, Python*

- Trained and fine-tuned CNN models on agricultural image datasets to classify cotton plant diseases, optimizing preprocessing pipelines with OpenCV for improved model accuracy.
- Built an end-to-end image classification pipeline from data augmentation to inference, enabling early disease prediction to support agricultural decision-making.

**Additional Projects:** Voice Assistant (Python) — desktop automation tool using speech recognition and TTS for app launching, web search, and reminders; Face Recognition System (Python, OpenCV) — real-time webcam-based face detection and recognition pipeline.

## Publication

*Machine Learning-Based Item Recommendation for Flipkart Using SVD Technique*  
Presented at the 2nd IEEE International Conference on New Frontiers in Communication, Automation, Management and Security (NFCAMS 2025)

## Hackathons, Certifications & Volunteering

- CodeFest '25 — 36-Hour Hackathon — Ultron 8.0 Hackvengers — 24-Hour Hackathon
- NPTEL: Java Programming — Introduction to Prompt Engineering with GitHub Copilot
- Community Connect, Ekta Shakti Foundation — Conducted basic computer literacy classes and assisted lab sessions for government school students.

## Soft Skills

Communication • Teamwork • Problem Solving • Adaptability • Cross-functional Collaboration